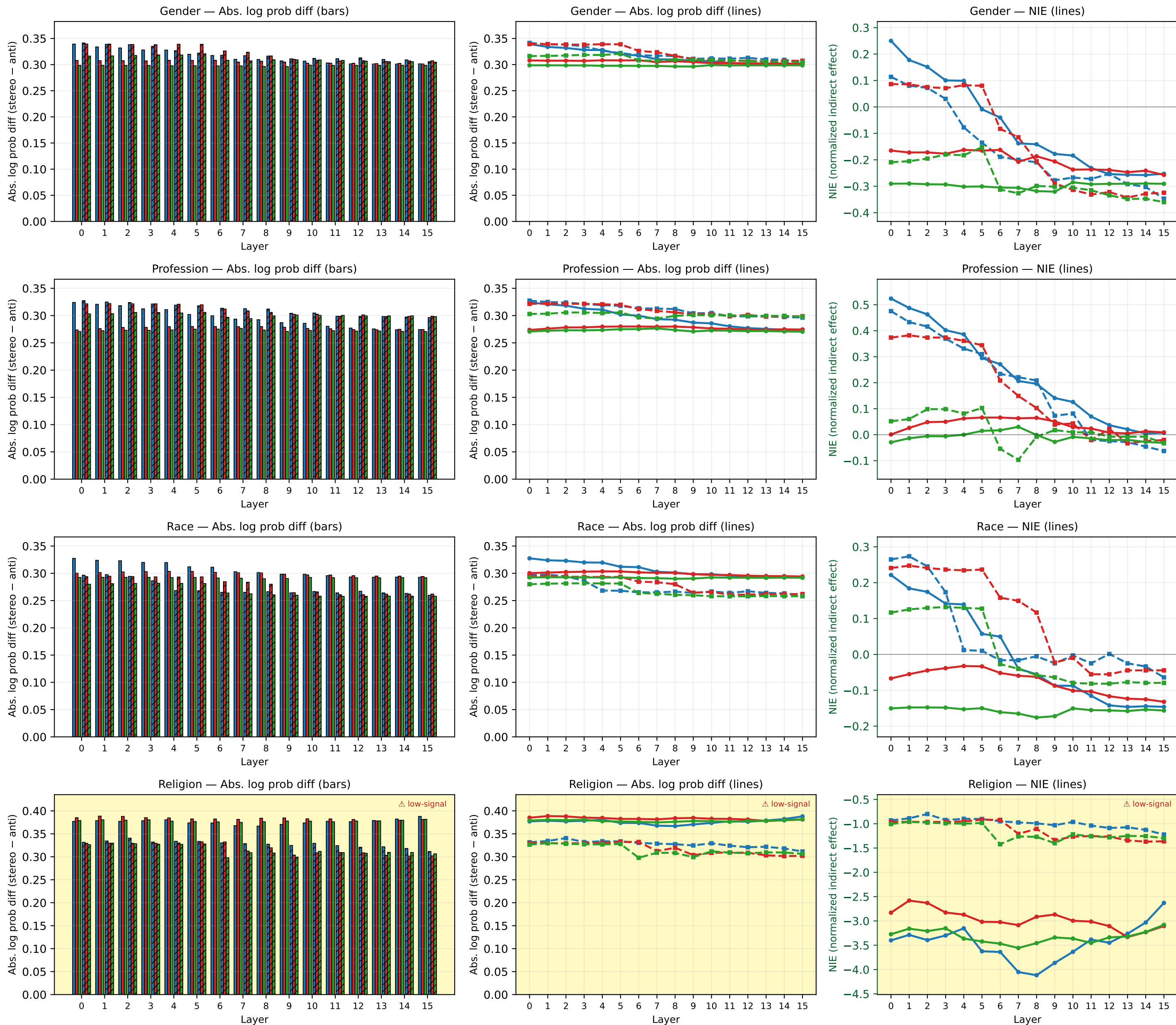


**OLMo-2-0425-1B (s2-51B) vs Pythia-1B (step143k) — cross-architecture pre-training comparison**  
**Panel 1: Abs. log prob diff (bars) | Panel 2: Abs. log prob diff (lines) | Panel 3: NIE (independent y-axis)**  
**Solid = OLMo, hatched/dashed = Pythia. Panels 1 & 2 share y-axis.**



Panel 1: solid=OLMo, hatched=Pythia | Panels 2-3: solid=OLMo, dashed=Pythia | #005C27 axis = independent NIE scale

|                     |                       |                     |                       |
|---------------------|-----------------------|---------------------|-----------------------|
| OLMo - States       | Pythia - Attn severed | OLMo - States       | Pythia - Attn severed |
| Pythia - States     | OLMo - MLP severed    | Pythia - States     | OLMo - MLP severed    |
| OLMo - Attn severed | Pythia - MLP severed  | OLMo - Attn severed | Pythia - MLP severed  |